The Diamond Toy Model for the Transport Invariant

A Proposed Paper AB for the Dream-Paper Suite

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March 20, 2026

Abstract

Paper AA defines the category **POT** and introduces the compression-ledger transport invariant $I(R,U) = \log|O_R(U)| + \lambda\,\Lambda_R(U) + \mu\,\log|C_R(U)|$. It identifies the Overlap Route-Agreement Lemma as the first forcing theorem and the uniqueness of the normalization coefficients (λ, μ, ν) as a key open obligation.

The present paper, Paper AB, provides the first concrete hand-computable testbed for these obligations. It constructs two four-node diamond diagrams of admissible enlargements, computes the transport invariant on each route, tests for route-invariance and composition- additivity, and identifies the first explicit algebraic obstruction to uniqueness of λ. It then introduces the collar-sensitive second diamond and uses it to distinguish the "missing boundary term" failure mode from the "genuinely nonlinear defect-compression interaction" failure mode.

The paper is a computational and conceptual foundation for the abstract theorems of Paper AA. No new mathematics beyond explicit finite computation is claimed. Its role is to convert the NFC Push program's iterative analysis into a fixed, precisely stated, referee- auditable finite laboratory.

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1 Purpose and design philosophy

Paper AA defines the transport invariant $I(R, U)$ and states the Overlap Route-Agreement Lemma as a target theorem. But it does not exhibit any concrete instance where the invariant can be computed by hand, route-invariance can be tested, or the failure of uniqueness can be observed and diagnosed.

The present paper fills that gap. Its purpose is twofold:

1. to exhibit the smallest finite diagrams on which the transport invariant is computable and route-invariance is testable; and
2. to identify, through explicit computation, which of the following failure modes appears first:
   (a) *Missing boundary term*: the naive invariant $\log|\Sigma| + \lambda\Lambda$ fails, but the boundary-corrected $\log|O| + \lambda\Lambda + \mu \log|C|$ succeeds;
   (b) *Nonlinear defect-compression interaction*: failure persists even after adding the boundary term, requiring the bilinear correction $\nu\Lambda \cdot \log|C|$.

The design philosophy is exactly that of the NFC Push program: start with the smallest possible model, hold all variables fixed except the one being tested, and diagnose failures before escalating.

2 Diamond 1: the collar-trivial control case

2.1 Setup

Diamond 1 is the smallest four-node diagram with two distinct admissible enlargement routes and a non-trivial interaction between compression and defect events.

Definition 2.1 (Diamond 1 regimes)**.** The four regimes of Diamond 1 are:

$$R_0 \longrightarrow R_A \longrightarrow R_2, \qquad R_0 \longrightarrow R_B \longrightarrow R_2,$$

with the following data drawn from the licensed **POT**-fiber structure:

Regime	$\lvert\Sigma\rvert$	$\lvert O\rvert$	$\lvert C\rvert$	Λ (cumulative)
R_0	3	3	2	0
R_A	2	2	1	0
R_B	3	3	2	1
R_2	2	2	1	1

The step types are:

- $R_0 \to R_A$: *right-stable merge* — two stabilized classes collapse, quotient size drops $3 \to 2$, collar collapses $2 \to 1$, no defect increment ($\Delta\Lambda = 0$);

- $R_A \to R_2$: *defect-mediated step* — quotient size unchanged ($2 \to 2$), collar already trivial ($1 \to 1$), defect increment $\Delta\Lambda = 1$;
- $R_0 \to R_B$: *defect-mediated step* — quotient size unchanged ($3 \to 3$), collar unchanged ($2 \to 2$), defect increment $\Delta\Lambda = 1$;
- $R_B \to R_2$: *right-stable merge* — quotient size drops $3 \to 2$, collar collapses $2 \to 1$, no defect increment ($\Delta\Lambda = 0$).

Remark 2.2 (Why this is the minimal diamond). A chain (no branching) cannot test route-invariance. A diamond where all steps are right-stable gives $\Lambda = 0$ everywhere, making the λ-term inert. A diamond where all steps are defect-mediated gives no right-stable compression, making the $\log|\Sigma|$-term constant. The chosen diamond is the smallest one where both step types appear, on both routes, in opposite orders.

2.2 Computation of I on Diamond 1

Computation 2.3 (Route A on Diamond 1). Using $I(R_0) = \log 3 + \lambda \cdot 0 + \mu \log 2 = \log 3 + \mu \log 2$:

$$\begin{aligned} I(R_A) &= \log 2 + \lambda \cdot 0 + \mu \log 1 = \log 2, \\ I(R_2) &= \log 2 + \lambda \cdot 1 + \mu \log 1 = \log 2 + \lambda. \end{aligned}$$

Total change along Route A:

$$\Delta I_A \;=\; I(R_2) - I(R_0) \;=\; \log 2 + \lambda - \log 3 - \mu \log 2 \;=\; \log(2/3) + \lambda - \mu \log 2.$$

Computation 2.4 (Route B on Diamond 1).

$$\begin{aligned} I(R_B) &= \log 3 + \lambda \cdot 1 + \mu \log 2 = \log 3 + \lambda + \mu \log 2, \\ I(R_2) &= \log 2 + \lambda \cdot 1 + \mu \log 1 = \log 2 + \lambda. \end{aligned}$$

Total change along Route B:

$$\Delta I_B \;=\; I(R_2) - I(R_0) \;=\; \log 2 + \lambda - \log 3 - \mu \log 2 \;=\; \log(2/3) + \lambda - \mu \log 2.$$

Proposition 2.5 (Route-invariance on Diamond 1)**.** *For the boundary-corrected invariant $I = \log|O| + \lambda\Lambda + \mu \log|C|$, the total change along Route A equals the total change along Route B on Diamond 1, for all values of λ and μ.*

Proof. By Computations 2.3 and 2.4, $\Delta I_A = \Delta I_B = \log(2/3) + \lambda - \mu \log 2$. □

Remark 2.6 (Why this does not yet determine λ). Proposition 2.5 shows Diamond 1 is consistent with any (λ, μ). It is a coherence test, not a uniqueness test. Uniqueness of the normalization requires a second diamond with a different quotient-drop profile, so that the same (λ, μ) must reconcile two structurally different cases simultaneously.

3 Diamond 2: the collar-sensitive diagnostic case

3.1 Setup

Diamond 2 holds the quotient data fixed at the Diamond 1 pattern but introduces a collar asymmetry on the intermediate node of Route B.

Definition 3.1 (Diamond 2 regimes).

Regime	$\lvert\Sigma\rvert$	$\lvert O\rvert$	$\lvert C\rvert$	Λ (cumulative)
S_0	3	3	2	0
S_A	2	2	1	0
S_B	3	3	**3**	1
S_2	2	2	1	1

The only change from Diamond 1 is $|C_{S_B}| = 3$ instead of 2.

3.2 Computation on Diamond 2

Computation 3.2 (Route A on Diamond 2). Identical to Diamond 1 (collar on Route A is trivial throughout):

$$\Delta I_A \;=\; \log(2/3) + \lambda - \mu \log 2.$$

Computation 3.3 (Route B on Diamond 2).

$$\begin{aligned} I(S_B) &= \log 3 + \lambda \cdot 1 + \mu \log 3 = \log 3 + \lambda + \mu \log 3, \\ I(S_2) &= \log 2 + \lambda + \mu \log 1 = \log 2 + \lambda. \end{aligned}$$

Total change along Route B:

$$\Delta I_B \;=\; \log 2 + \lambda - \log 3 - \mu \log 3 \;=\; \log(2/3) + \lambda - \mu \log 3.$$

Proposition 3.4 (Route-invariance failure on Diamond 2). *For the boundary-corrected invariant* $I = \log|O| + \lambda\Lambda + \mu \log|C|$*:*

$$\Delta I_A \;=\; \log(2/3) + \lambda - \mu \log 2, \qquad \Delta I_B \;=\; \log(2/3) + \lambda - \mu \log 3.$$

These are equal if and only if $\mu \log 2 = \mu \log 3$*, which requires* $\mu = 0$*.*

Remark 3.5 (What the failure means). Setting $\mu = 0$ recovers the naive invariant $\log|\Sigma| + \lambda\Lambda$, which by Proposition 2.5 is route-invariant on Diamond 1 for any λ. So the boundary-corrected invariant, if $\mu \neq 0$, fails on Diamond 2. This is *not* a defect of the invariant; it is a signal. It says the defect event on Route B, which occurs inside a larger realized collar sector ($|C| = 3$ vs. $|C| = 2$ on Diamond 1), carries a different effective burden than the same defect event on Route A, where the collar is trivial. That is the obstruction the bilinear term is designed to absorb.

4 The bilinear correction and the joint normalization

4.1 Introducing the bilinear term

Definition 4.1 (Nonlinear invariant on the diamonds).

$$I_{\mathrm{nl}}(R, U) \;=\; \log|O| \;+\; \lambda\,\Lambda \;+\; \mu\,\log|C| \;+\; \nu\,\Lambda \cdot \log|C|.$$

Computation 4.2 (Route A on Diamond 2 under I_{nl}). On Route A, the defect step $R_A \to R_2$ occurs at $|C_{R_A}| = 1$:

$$\text{bilinear contribution at defect step:} \quad \nu \cdot 1 \cdot \log 1 = 0.$$

So $\Delta I_{\mathrm{nl}A} = \log(2/3) + \lambda - \mu \log 2$.

Computation 4.3 (Route B on Diamond 2 under I_{nl}). On Route B, the defect step $S_0 \to S_B$ occurs at $|C_{S_0}| = 2$ (source collar convention):

$$\text{bilinear contribution at defect step:} \quad \nu \cdot 1 \cdot \log 2.$$

So

$$I_{\mathrm{nl}}(S_B) = \log 3 + \lambda + \mu \log 3 + \nu \cdot \log 2,$$

and the total change along Route B is:

$$\Delta I_{\mathrm{nl}B} = \log(2/3) + \lambda - \mu \log 3 + \nu \log 2.$$

Proposition 4.4 (Joint route-invariance under I_{nl})**.** *The nonlinear invariant I_{nl} is route-invariant on both diamonds simultaneously if and only if:*

$$\textit{Diamond 1:} \quad \mu \log 2 = \mu \log 2 \quad (\textit{trivially satisfied}), \tag{1}$$

$$\textit{Diamond 2:} \quad \mu \log 2 = \mu \log 3 - \nu \log 2. \tag{2}$$

Equation (2) gives:

$$\nu \;=\; \frac{\mu(\log 3 - \log 2)}{\log 2} \;=\; \mu\,\frac{\log(3/2)}{\log 2}.$$

In particular, for any fixed $\mu > 0$, the bilinear coefficient ν is uniquely determined by joint route-invariance.

Proof. Setting $\Delta I_{\mathrm{nl}A} = \Delta I_{\mathrm{nl}B}$ on each diamond yields the stated equations. Solving (2) for ν gives the formula. λ remains free at this stage, fixed only by the composition-additivity condition (Step 3 of Paper AA's theorem ladder). □

Remark 4.5 (Significance of Proposition 4.4). This is the first concrete instance of uniqueness in the normalization program. Given μ, the ratio ν/μ is forced by route-invariance alone, without any phenomenological input. The bilinear term is not arbitrary flexibility; it is the unique correction needed to absorb the collar-context sensitivity of defect events. This is the diagnostic result the NFC Push program predicted: if the boundary-corrected linear invariant fails on the second diamond, the bilinear term has a clean explanatory job to do.

5 The Defect-in-Collar Amplification Lemma

The computations above point toward an abstract structural principle.

Definition 5.1 (Amplification functional)**.** For an admissible regime R and local region U, the *defect amplification functional* is

$$A_R(U) \;=\; \nu_{\mathrm{norm}} \cdot \Lambda_R(U) \cdot \log |C_R|(U),$$

where ν_{norm} is the normalization coefficient forced by joint route-invariance on the diamond pair.

Lemma 5.2 (Defect-in-Collar Amplification Lemma — toy instance)**.** *On the diamond pair of Definitions 2.1 and 3.1, the amplification functional $A_R(U)$ satisfies:*

1. *$A_R(U) = 0$ whenever $|C_R(U)| = 1$ (collar-trivial regime);*
2. *$A_R(U) > 0$ whenever $|C_R(U)| > 1$ and $\Lambda_R(U) > 0$;*

3. *the effective transported burden of a defect step is* $\Delta_{\text{eff}} = \Delta\Lambda + A_R(U)$*, and this is route-invariant on both diamonds simultaneously.*

Proof. Property (1) follows from $\log 1 = 0$. Property (2) is immediate. Property (3) follows from Proposition 4.4 with $\nu = \mu \cdot \log(3/2)/\log 2$. □

Remark 5.3 (Status of the abstract amplification lemma). Lemma 5.2 is a proof of the amplification principle on the specific diamond pair. The abstract Defect-in-Collar Amplification Lemma of Paper AA (AA Step 5) requires the same statement on all admissible diamond diagrams over all regimes. That general statement remains open. The present toy instance is the first positive evidence that the principle is not merely plausible but concretely realized.

6 The Amplified Minimal-Carrier Theorem in the toy setting

Definition 6.1 (Carrier families on Diamond 2)**.** A *carrier family* on Diamond 2 is an assignment $\mathcal{K} = (K_{S_0}, K_{S_A}, K_{S_B}, K_{S_2})$ of realized outcome sets, one for each regime node, such that:

1. $K_R \subseteq O_R$ at each node;
2. the transition maps restrict to $\mathcal{K}$; and
3. the amplified invariant I_{nl} factors through $\mathcal{K}$.

The *minimal carrier* $\mathcal{K}^{\min}$ is the carrier family that carries exactly the observed outcomes in Definition 3.1 and no more.

Proposition 6.2 (Minimal-carrier uniqueness on Diamond 2)**.** *Under* I_{nl} *with the normalization of Proposition 4.4, the minimal carrier* $\mathcal{K}^{\min}$ *is the unique carrier family through which* I_{nl} *factors with minimal total burden. Any strictly larger carrier* $\mathcal{K}^{\text{fat}}$ *carrying an extra collar-sensitive distinction on* S_B *has a strictly larger value of* $I_{\text{nl}}(S_B)$ *and is therefore not minimal.*

Proof. Any extra outcome class on S_B increases $|O_{S_B}|$ and hence $\log |O_{S_B}|$. Since the amplified invariant is minimized by the smallest realized outcome set consistent with factorization, $\mathcal{K}^{\min}$ is the unique minimizer. □

Remark 6.3 (What this demonstrates). Proposition 6.2 is the toy-level instance of the Amplified Minimal-Carrier Theorem. It shows that the amplified invariant, once properly normalized, distinguishes a minimal carrier from a redundant one on the specific diamond pair. It does not yet prove the general theorem (which requires the abstract route-agreement and coherent-transition-map results), but it exhibits the mechanism concretely.

7 The partial order on carrier families

Definition 7.1 (Realization-support order)**.** For carrier families $\mathcal{K}$ and $\mathcal{K}'$ on a diamond diagram, write $\mathcal{K} \leq \mathcal{K}'$ if:

1. $K_R \subseteq K'_R$ for every regime node R (inclusion of realized outcome sets);
2. $|C_R(K)| \leq |C_R(K')|$ for every R (collar support is no larger); and
3. the defect-ledger increments of $\mathcal{K}$ are dominated by those of $\mathcal{K}'$.

This is a partial order on carrier families, and the minimal carrier of Definition 6.1 is the least element under this order among all carrier families through which I_{nl} factors faithfully.

Remark 7.2 (Why route-agreement and realized-type discipline are enforced). The partial order in Definition 7.1 uses only realized stabilized outcome and collar data, not raw token counts. This respects the suite's core discipline that only realized stabilized objects are licensed as objective. The order is well-defined on **POT**-fibers because the fiber data already carries realized, not raw, collar and outcome information.

8 Summary of the diamond testbed

The following table summarizes what the two-diamond pair achieves as a testbed.

Test	Result	Lesson
Route-invariance of naive $\log\lvert\Sigma\rvert + \lambda\Lambda$ on Diamond 1	Pass (for any λ)	Consistent but doesn't fix λ
Route-invariance of $\log\lvert O\rvert + \lambda\Lambda + \mu\log\lvert C\rvert$ on Diamond 1	Pass (for any λ, μ)	Boundary term adds no information here
Route-invariance of $\log\lvert O\rvert + \lambda\Lambda + \mu\log\lvert C\rvert$ on Diamond 2	Fail (unless $\mu = 0$)	Collar-context sensitivity is real
Route-invariance of I_{nl} on Diamond 2	Pass iff $\nu = \mu \cdot \log(3/2)/\log 2$	Bilinear term absorbs obstruction; ν/μ uniquely fixed
Minimal carrier uniqueness under I_{nl}	Pass	Amplified invariant selects $\mathcal{K}^{\min}$

9 Status ledger

Label	Claim	Status
AB1	Route-invariance of I_{nl} on Diamond pair (toy instance)	Unc
AB2	Defect-in-Collar Amplification Lemma on diamond pair	Unc
AB3	Uniqueness of ν/μ from joint route-invariance on the diamond pair	Unc
AB4	Minimal-carrier uniqueness under I_{nl} on Diamond 2	Unc
AB5	Abstract route-agreement on all admissible diagrams	Open (AA Step 1)
AB6	Abstract uniqueness of (λ, μ, ν) on all admissible regimes	Open (AA Step 4)
AB7	Amplified Minimal-Carrier Theorem (general)	Open (AA Step 5)

10 Conclusion

Paper AB establishes the two-diamond pair as the suite's canonical toy testbed for the transport invariant. The key results are:

1. The naive invariant $\log|\Sigma| + \lambda\Lambda$ is route-invariant on Diamond 1 but provides no uniqueness information.
2. The boundary-corrected invariant $\log|O| + \lambda\Lambda + \mu\log|C|$ fails on Diamond 2 unless $\mu = 0$, diagnosing real collar-context sensitivity.
3. The bilinear correction $\nu\Lambda \cdot \log|C|$ repairs this failure and forces the ratio $\nu/\mu = \log(3/2)/\log 2$ uniquely by route-invariance.
4. The amplified invariant selects a unique minimal carrier on Diamond 2.

These results are the first positive evidence for the abstract theorem program of Paper AA. They show that the transport invariant is not a philosophical proposal: it is a concrete quantity that behaves exactly as the NFC Push program predicted, in a finite hand-computable laboratory built entirely from **POT**-licensed data.

The next step is the abstract version: prove the Overlap Route-Agreement Lemma for all admissible overlaps, not just the diamond toy. That is the route from this toy model to the general theorem.